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Curating for Efficiency: A Blueprint for Building Specialized LLMs on Low-Power Hardware

Source Identification and Provenance for Domain-Specific Datasets

The foundation of any successful LLM training project lies in the selection of appropriate, high-quality source materials. For resource-constrained environments, this initial step is critical, as starting with well-curated, permissively licensed, and often already-cleaned datasets can significantly reduce downstream computational overhead. The approach to sourcing varies dramatically by domain, requiring a tailored search for foundational corpora that align with the goals of efficiency and quality. Across all domains, the principles of data provenance, transparency, and traceability are paramount [70]. Understanding where data originates, how it was processed, and what licenses govern its use is not merely an academic exercise but a practical necessity for legal compliance and ethical AI development [99] [112].

For a dataset focused on **General Knowledge**, the primary challenge is balancing scale with quality and legality. Massive collections of text form the basis of modern LLM pre-training [150], but raw web-scraped data is often noisy and carries complex licensing issues [39]. To address this, large-scale ethical datasets have been constructed specifically to provide a foundation for open and auditable models. The `Common Pile v0.1` stands out as a premier resource, comprising an eight-terabyte collection of public domain and openly licensed text designed explicitly for LLM pre-training [40] [97]. Its creators aimed to support the development of fully open models by ensuring the legal right to release the source material is clear [27]. Similarly, the `Common Corpus` provides another massive, ethically sourced dataset intended to facilitate pre-training [30]. These resources aggregate data from diverse origins while attempting to manage licensing complexities, a crucial factor for any project aiming for broad accessibility [41]. The provenance of such datasets is a subject of active research, with frameworks emerging to evaluate dataset compliance with governance plans and provide traceability back to original sources [130] [131].

In the domain of **Technical Documentation**, the objective shifts from general world knowledge to specialized, structured information. A significant finding is the existence of dedicated resources built for this purpose. The `propella-1` dataset is a notable example, consisting of small multilingual LLMs trained on documents annotated across 18 different properties, indicating a sophisticated pipeline exists to transform raw documentation into a structured, training-ready format [23]. Beyond leveraging existing datasets, a parallel strategy involves optimizing common documentation formats like HTML and PDF for LLM consumption. Research shows that converting these formats to plain text can introduce noise and irrelevant tokens, degrading model performance [157]. Therefore, using tools for efficient HTML-to-Markdown conversion [53] or employing heuristic-based chunking techniques for PDFs [52] is essential. This ensures that the rich semantic information within documentation is preserved without adding unnecessary computational burden during training. Continued pre-training on such specialized documentation has been shown to improve an LLM's ability to perform domain-specific tasks accurately [83].

For **Conversational Dialogue**, the focus is on capturing natural language patterns, context, and instruction-following capabilities. The `LMSYS-Chat-1M` dataset is a state-of-the-art resource in this area, containing one million real-world conversations involving 25 state-of-the-art LLMs [42]. Its creation involved careful filtering to ensure quality, diversity, and relevance for alignment tasks. Another relevant resource is the Chinese Open Instruction Generalist (COIG) corpus, which aims to maintain a harmless, useful, and diverse collection of teaching data [151]. The curation of conversation data presents unique challenges related to maintaining naturalness and avoiding biases present in the source material. The provenance of user-generated conversational data also necessitates careful attention to privacy and consent protocols [83]. Given the potential for noise and unstructured nature of conversation logs, leveraging a pre-filtered, high-quality dataset like `LMSYS-Chat-1M` is a highly efficient strategy, allowing the focus to shift directly to fine-tuning rather than building a new corpus from scratch.

Finally, for **Code Generation**, the need for syntactic correctness and semantic clarity elevates data quality to a higher level of importance. The BigCode project has produced definitive resources for this domain. The `StarCoder2` models were trained on `The Stack v2`, a dataset of 783GB of code spanning 86 programming languages [103]. This dataset was constructed with a strong emphasis on responsible development, including meticulous license filtering, malware removal, and validation to ensure the permissive licensing of its contents [120][122][135]. Other specialized code datasets include `SwallowCode`, a high-quality Python dataset derived from `The Stack v2`, and `ShortCoder`, which focuses on syntax optimization [100][134]. The provenance of code data is particularly critical due to the legal and security implications of copyright infringement and malicious software. The BigCode project's collaboration with Software Heritage to build upon a verified archive of source code exemplifies best practices in this area [122].

Domain	Recommended High-Quality Dataset(s)	Key Characteristics & Provenance
General Knowledge	Common Pile v0.1 [40] [97], Common Corpus [30]	8TB+ of public domain and openly licensed text; ethically sourced to support open and auditable models [27].
Technical Documentation	propella-1 [23], Optimized Documentation Formats (HTML/PDF)	Trained on annotated documents across 18 properties [23]; requires efficient conversion to plain text to minimize noise [52] [53].
Conversational Dialogue	LMSYS-Chat-1M [42], COIG [151]	1 Million real-world conversations for alignment; COIG focuses on creating a diverse, harmless, and useful instruction corpus [151].
Code Generation	The Stack v2 [103], StarCoder2 Models [102]	783GB of code in 86 languages with rigorous license filtering and malware removal [103][135]; built upon Software Heritage archive [122].

Curation Strategies for Computational Efficiency

Once a high-quality source dataset has been selected, the next phase is curation—the systematic process of cleaning, filtering, and structuring the data to maximize its utility for training while minimizing computational cost. In a low-CPU environment, the efficiency of the curation pipeline itself becomes a primary concern. Standard data preprocessing pipelines often run on the CPU, which can become a bottleneck when its processing speed is lower than the rate at which the GPU can train, leading to idle hardware and wasted time [44] [79]. Therefore, the strategic goal is to design a "lightweight, purpose-driven data pipeline" that prioritizes speed and simplicity without sacrificing the core quality attributes of the dataset [16]. This involves making deliberate choices about filtering techniques, deduplication methods, and overall data recipe design.

A central concept in this process is the "data recipe," which encompasses the entire data processing pipeline transforming raw sources into a final training corpus [18]. Research has shown that testing various recipes, which involve different combinations of domain composition, quality filtering, and deduplication strategies, is crucial for identifying the most effective approach for a given task [20]. For a resource-constrained setup, the ideal recipe would favor computationally inexpensive operations. License validation is a prime example; since many datasets are composed of permissively licensed content, a simple filter based on known license lists can be applied efficiently [39] [138]. Similarly, basic text cleaning and normalization steps, such as removing extraneous characters or standardizing whitespace, are essential first steps that do not require significant computational power [9].

For the domain of **code generation**, data quality is non-negotiable. A major source of noise and incorrect learning signals in code datasets is syntax errors. Advanced curation pipelines have therefore incorporated linter-based filtering to systematically remove code snippets with syntactic issues, significantly improving the final quality of the training data [11]. Furthermore, research has demonstrated that simply curating existing data is not always optimal; one can actively *improve* it. Model-agnostic simplification solutions like `SlimCode` and `LEANCODE` apply transformation rules to make code more readable and less complex, which has been shown to improve LLM performance on downstream code tasks [133][148][149]. This approach enhances the signal-to-noise ratio in the data, allowing the model to learn more effectively. The role of verification tests in guiding the model towards correct solutions is also critical, highlighting the importance of including well-structured problem-solution pairs in the dataset [107][154].

In the case of **technical documentation** and other text-heavy domains, a significant portion of the curation effort may involve handling unstructured formats like HTML or PDF. Converting these to clean, plain text is a necessary preprocessing step, but it must be done efficiently. Using optimized libraries for HTML-to-Markdown conversion can preserve structural elements without introducing the parsing overhead of a full browser engine [53]. For PDFs, novel heuristic-based chunking techniques like ChunkNorris have been developed to intelligently break down documents into semantically coherent chunks, avoiding the noise that comes from naive, fixed-size splitting [52]. These intelligent chunking strategies are vital for Retrieval-Augmented Generation (RAG) systems and direct training, as they ensure the model learns from meaningful segments of text [139].

Across all domains, deduplication is a critical curation step to prevent the model from overfitting to repeated examples. While powerful GPU-accelerated frameworks for deduplication exist, their underlying algorithms can be adapted for CPU-based execution [28] [58]. MinHash Locality-Sensitive Hashing (LSH), for instance, is a computationally efficient technique for identifying near-duplicate documents that can be implemented in a CPU-friendly manner. The curation process can also leverage LLMs themselves, though this must be approached with caution in a low-CPU setting. LLM-based rating systems can offer a cost-effective alternative to human annotation for scoring data quality, but they are prone to inaccuracies and biases [17]. A hybrid approach, where a lightweight rule-based system performs initial filtering and a more powerful (but potentially offline) LLM is used for nuanced evaluation, could be a viable compromise. Ultimately, the most efficient strategy is to leverage existing, already-cleansed datasets whenever possible, reserving intensive curation efforts for creating smaller, specialized subsets from larger corpora.

Optimizing Data Structure and Tokenization for Low-CPU Environments

Beyond the efficiency of the curation pipeline, the intrinsic structure of the final dataset and the method used for tokenization play a decisive role in determining the computational load during model training. The CPU is heavily involved in these stages, and inefficiencies here can create significant bottlenecks even before the data reaches the GPU for processing [6]. An effective strategy for low-CPU environments involves designing the dataset to be as compact and semantically dense as possible, while also selecting a tokenizer that minimizes the number of tokens required to represent the information, thereby reducing both memory usage and computational complexity.

Tokenization is the process of breaking down text into discrete units called tokens, which are the fundamental inputs for an LLM [13]. The choice of tokenizer and its vocabulary size directly impacts model performance and efficiency. Effective tokenizers enhance a model's understanding of context and semantics, which can lead to improved performance with fewer tokens [8]. For instance, Byte Pair Encoding (BPE) is a common algorithm used in many popular tokenizers. However, for specialized domains like code, domain-specific tokenizers or tokenization strategies can yield better results. Research into token-efficient notations like ONTO (Object Notation for Token Optimization) demonstrates a pathway toward more efficient input representations by using a columnar format that declares field names once, reducing redundancy [110]. Simpler, yet effective, strategies include ensuring that the input data is clean and free of superfluous characters or formatting that would otherwise be converted into additional tokens.

Data structure is equally important. For **conversational dialogue**, long, unbroken transcripts can be challenging for an LLM to process. A common practice is to split dialogues into individual turns or messages, creating a sequence of user queries and model responses. This structured format is more amenable to supervised fine-tuning and helps the model learn the conversational flow. For **technical documentation**, structuring the data into well-defined chunks, each representing a single function, class, or API endpoint, is highly beneficial. Intelligent chunking strategies that aim for semantic coherence rather than arbitrary length are superior for retrieval and training tasks [139]. In the context of **code generation**, the inherent structure of the language can be leveraged. Instead of feeding the model massive, monolithic files, breaking them down into smaller, self-contained functions or code blocks is a standard and effective practice [50]. This not only makes the training process more manageable but also encourages the model to learn modular coding patterns.

The interplay between data structure and model architecture is also a key consideration. For example, some training methodologies construct samples by randomly sampling documents and separating them with special beginning-of-sequence (BOS) and end-of-sequence (EOS) tokens [140]. While this can increase data diversity, it may not be the most efficient approach. Structured packing, which organizes documents within a single training sample based on their relationships, can improve how the model utilizes long context windows [140]. The choice of data format for storage also matters. While plain text is simple, binary formats like Parquet can offer more efficient storage and faster reading speeds, reducing the I/O bottleneck on the CPU.

Ultimately, the goal is to create a dataset that is "token-efficient." This means every token in the input sequence provides maximum semantic value to the model. Techniques that improve code readability, such as those proposed by SlimCode and LEANCODE, contribute to this goal by removing syntactic noise that does not add to the functional meaning of the code [133][149]. By combining a clean, well-structured data layout with an efficient tokenization strategy, the total number of tokens processed during training can be minimized. This reduction directly translates to lower computational requirements, shorter training times, and a better fit for the constraints of a low-speed CPU environment.

Training Methodologies Aligned with Resource Constraints

The ultimate objective of dataset creation is to train a capable LLM, but the user's constraint of low CPU speed necessitates a departure from conventional, resource-intensive training paradigms. Full-scale pre-training of a large language model from scratch typically requires thousands of high-end GPUs and vast amounts of data, making it impractical for most users [128]. A more viable and strategically sound approach for a resource-constrained environment is to leverage a pre-trained base model and adapt it to the target domain using either continued pre-training or fine-tuning. These methods allow the model to build upon its existing general knowledge while specializing it with the high-quality, single-domain data created through the curation process.

Continued pre-training involves extending the initial pre-training phase of a model by exposing it to a new, specialized dataset [117]. This process is distinct from fine-tuning because the model's weights are updated across all layers using the new data. It is particularly effective for adapting a model to a new language or domain, as demonstrated by the use of a ~32GB medical dataset to enhance a model's knowledge in that area [115] [116]. A step-by-step guide for continuous pre-training highlights its feasibility for language adaptation, showing how a model like Poro 2's performance in Finnish was boosted using Llama 3.1 [115]. This approach is well-suited for the user's plan, as the curated single-domain datasets (e.g., a 783GB code corpus or a 3TB technical documentation set) can serve as the extended pre-training data. This allows the model to learn the specific vocabulary, syntax, and concepts of the target domain deeply.

Fine-tuning, on the other hand, involves taking a pre-trained model and continuing to train it on a smaller, task-specific dataset, often with a lower learning rate [76]. This is the standard approach for aligning models with specific instructions or behaviors, such as improving code review accuracy [77] or enhancing performance in a particular application domain [83]. Fine-tuning is generally less computationally expensive than continued pre-training because it operates on a model that has already learned fundamental linguistic representations from a much larger, general corpus [31]. Supervised fine-tuning on a unified, high-quality dataset has been shown to yield significant performance gains, around 20% in one study [95]. Given that the user is creating curated datasets for specific domains, fine-tuning is a highly appropriate strategy. It enables the specialization of a powerful base model (like Gemma or Llama) with minimal data compared to the terabytes required for initial pre-training.

Regardless of the chosen method, the design of the dataset must align with the training strategy. For instance, if the goal is to train an agent, the dataset might need to contain multi-turn interactions or reasoning traces [67] [68]. If the goal is to improve factual accuracy, the dataset should be rich in verifiable facts and ideally include structures that support verification, such as links to knowledge graphs [35] [111]. The choice of optimization techniques during training is also influenced by the target deployment environment. Since the final model will likely run on low-power hardware like a Raspberry Pi, quantization techniques that reduce the precision of the model's weights (e.g., to INT8 or INT4) should be considered [43] [94]. Quantization can dramatically reduce the model's memory footprint and accelerate inference, making it feasible to run on CPUs [45]. Some training processes now incorporate automatic quantization flows, further streamlining this optimization step [94]. By selecting a base model that is compatible with these optimizations and designing the training process accordingly, the resulting LLM will be not only knowledgeable in its specialized domain but also efficient enough for deployment in the target low-CPU environment.

Synthesizing a Strategic Framework for Dataset Development

Synthesizing the analysis of source materials, curation strategies, data optimization, and training methodologies provides a comprehensive, actionable framework for developing a high-quality, single-domain dataset for training an LLM under low CPU speed constraints. The core insight is that success hinges on resolving the inherent tension between data quality and computational efficiency not by compromising on one, but by strategically selecting resources and applying intelligent, lightweight techniques that favor efficiency without diluting quality. The following framework outlines a phased, evidence-based approach to achieve this goal.

The first strategic pillar is **Prioritizing Pre-Processed, Open-Source Datasets**. Given the computational limitations, the most direct path to a high-quality dataset is to start with existing, ethically sourced, and permissively licensed corpora. This avoids the immense upfront cost of scraping and initially cleaning vast quantities of raw data. For each domain, specific foundational datasets have emerged as standards: `The Stack v2` for code [103][135], `Common Pile` for general knowledge [97], `LMSYS-Chat-1M` for conversation [42], and resources like `propella-1` for technical documentation [23]. These datasets have already undergone significant curation, and their transparent provenance and licensing simplify compliance and ethical considerations [120][138].

The second pillar is adopting a **Two-Pronged Curation Strategy**. This involves separating computationally intensive operations from those performed during the actual training loop. Heavy filtering, such as complex license analysis, malware scanning, or deep semantic deduplication, should be performed offline as part of an initial, one-time curation pass. The output of this pass should be a smaller, refined dataset stored in an efficient format like JSONL or Parquet. During the training process itself, the preprocessing pipeline should be kept extremely lightweight, focusing only on essential tasks like tokenization and batching. This design directly addresses the known bottleneck where CPU-bound preprocessing slows down the overall training workflow [44] [79] .

The third pillar focuses on **Maximizing Data Efficiency**. This involves two key actions. First, actively improve the signal-to-noise ratio of the data. For code, this means applying model-agnostic simplification rules (`SlimCode`, `LEANCODE`) to make the code more readable and thus easier for the model to learn from [133][149]. Second, optimize the data's representation. This includes using token-efficient input notations where possible [110] and ensuring the data is structured in a way that maximizes semantic density, such as breaking down large documents or code files into meaningful, self-contained chunks [50] [139].

The fourth pillar dictates the adoption of an **Efficient Training Paradigm**. Full-scale pre-training is infeasible under the user's constraints. The recommended strategy is to take a publicly available, general-purpose base model (e.g., Gemma, Llama) and specialize it using either continued pre-training or fine-tuning. Continued pre-training is suitable for immersing the model in a new domain's vocabulary and style, while fine-tuning is more appropriate for aligning the model with specific tasks or instructions [76] [117]. Both methods are far more resource-efficient than training from scratch and are perfectly suited to the use of the curated single-domain datasets.

By integrating these four pillars, a robust and efficient workflow emerges. It begins with leveraging existing, high-quality datasets to save time and resources. It then applies a streamlined, two-phase curation process to ensure data quality without creating a CPU bottleneck. The curated data is structured and tokenized for maximum efficiency. Finally, this specialized data is used to adapt a powerful base model using a computationally tractable training methodology. This strategic framework directly addresses the user's research goal, providing a clear path to developing a specialized, high-performance LLM that is tailored to a specific domain and constrained by the realities of low CPU speed.

Reference

1. (PDF) Towards Best Practices for Open Datasets for LLM Training https://www.researchgate.net/publication/388068214_Towards_Best_Practices_for_Open_Datasets_for_LLM_Training

2. The Largest Collection of Ethical Data for LLM Pre-Training - arXiv https://arxiv.org/html/2506.01732v3
3. Conversation with Craig McLuckie on data provenance at All Things ... https://www.linkedin.com/posts/neilbagchi_allthingsopen-activity-7383578748937252864-3HGM
4. An Evaluation of LLMs Inference on Popular Single-board Computers https://arxiv.org/html/2511.07425v1
5. Run a Large Language Model (LLM) chatbot on a Raspberry Pi 5 https://learn.arm.com/learning-paths/embedded-and-microcontrollers/llama-python-cpu/llama-python-chatbot/
6. Characterizing CPU-Induced Slowdowns in Multi-GPU LLM Inference https://arxiv.org/html/2603.22774v1
7. [PDF] Efficient Continual Pre-training of LLMs for Low-resource Languages https://aclanthology.org/2025.naacl-industry.25.pdf
8. (PDF) Comparative Analysis of Tokenization Algorithms for Low ... https://www.researchgate.net/publication/391147271_Comparative_Analysis_of_Tokenization_Algorithms_for_Low-Resource_Language_Dzongkha
9. Data Preprocessing Techniques for LLMs | PDF | Learning - Scribd https://www.scribd.com/presentation/960503579/5-Data-Preprocessing-for-LLMs
10. Efficient LLMs Training and Inference: An Introduction - IEEE Xplore https://ieeexplore.ieee.org/iel8/6287639/6514899/10756602.pdf
11. Rewriting Pre-Training Data Boosts LLM Performance in Math and ... https://arxiv.org/html/2505.02881v1
12. Seed-Coder: Let the Code Model Curate Data for Itself - arXiv https://arxiv.org/html/2506.03524v2
13. A Comprehensive Overview of Large Language Models https://dl.acm.org/doi/10.1145/3744746
14. Mastering LLM Techniques: Text Data Processing - NVIDIA Developer https://developer.nvidia.com/blog/mastering-llm-techniques-data-preprocessing/
15. Efficient LLMs Training and Inference: An Introduction - IEEE Xplore https://ieeexplore.ieee.org/iel8/6287639/10820123/10756602.pdf
16. Lightweight, Purpose-driven Data Pipeline for Large Language ... https://arxiv.org/html/2411.11289v1
17. Improving Data Efficiency via Curating LLM-Driven Rating Systems https://arxiv.org/html/2410.10877v1

18. DataChef: Cooking Up Optimal Data Recipes for LLM Adaptation via ... https://arxiv.org/html/2602.11089v1
19. A Comparative Study of LLM Post-Training Data Quality and Model ... https://arxiv.org/html/2506.06522v3
20. Can Small Training Runs Reliably Guide Data Curation? Rethinking ... https://arxiv.org/html/2512.24503v1
21. MDCure: A Scalable Pipeline for Multi-Document Instruction-Following https://arxiv.org/html/2410.23463v2
22. Machine Learning Data Practices through a Data Curation Lens - arXiv https://arxiv.org/html/2405.02703v1
23. propella-1: Multi-Property Document Annotation for LLM Data ... - arXiv https://arxiv.org/html/2602.12414v1
24. DCA-Bench: A Benchmark for Dataset Curation Agents - arXiv https://arxiv.org/html/2406.07275v1
25. What Makes a High-Quality Training Dataset for Large Language ... https://dl.acm.org/doi/10.1145/3691620.3695061
26. PiKa: Expert-Level Synthetic Datasets for Post-Training Alignment ... https://arxiv.org/html/2510.06670v2
27. [PDF] THE LARGEST COLLECTION OF ETHICAL DATA FOR LLM PRE ... https://openreview.net/pdf?id=0wSlFpMsGb
28. FED: Fast and Efficient Dataset Deduplication Framework with GPU ... https://arxiv.org/html/2501.01046v3
29. [PDF] A Rigorous Evaluation of LLM Data Generation Strategies for Low ... https://aclanthology.org/2025.emnlp-main.418.pdf
30. Common Corpus: The Largest Collection of Ethical Data for LLM Pre ... https://openreview.net/forum?id=0wSlFpMsGb
31. Training Methods for Large Language Models: Current Approaches ... https://www.mdpi.com/2227-7080/14/2/133
32. Hallucination to Truth: A Review of Fact-Checking and Factuality ... https://arxiv.org/html/2508.03860
33. Hallucination to truth: a review of fact-checking and factuality ... https://link.springer.com/article/10.1007/s10462-025-11454-w
34. DefAn: Definitive Answer Dataset for LLM Hallucination Evaluation https://www.mdpi.com/2078-2489/16/11/937
35. LLM experimentation through knowledge graphs: Towards improved ... https://www.sciencedirect.com/science/article/pii/S1570826824000398

36. Evaluating Language Models for Biomedical Fact-Checking - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12458922/
37. Learning to Verify Summary Facts with Fine-Grained LLM Feedback https://aclanthology.org/2025.coling-main.16/
38. Unveiling Pitfalls and Potentials in Fact Verifiers | OpenReview https://openreview.net/forum?id=3NjnRo6apU
39. [PDF] Towards Best Practices for Open Datasets for LLM Training - arXiv https://arxiv.org/pdf/2501.08365
40. An 8TB Dataset of Public Domain and Openly Licensed Text - arXiv https://arxiv.org/html/2506.05209v1
41. A large-scale audit of dataset licensing and attribution in AI - Nature https://www.nature.com/articles/s42256-024-00878-8
42. LMSYS-Chat-1M: A Large-Scale Real-World LLM Conversation ... https://arxiv.org/html/2309.11998v4
43. LLMPi: Optimizing LLMs for High-Throughput on Raspberry Pi - arXiv https://arxiv.org/html/2504.02118v1
44. Data × LLM: From Principles to Practices - arXiv https://arxiv.org/html/2505.18458v1
45. [PDF] LiteRT-Optimized INT8 LLM for Raspberry Pi4 Deployment https://openaccess.thecvf.com/content/ICCV2025W/AIM/papers/Yoon_LiteRT-Optimized_INT8_LLM_for_Raspberry_Pi4_Deployment_ICCVW_2025_paper.pdf
46. Gemma 3 on Raspberry Pi 5: Real Benchmarks [2026] https://www.kunalganglani.com/blog/gemma-3-raspberry-pi-5-benchmark
47. Evaluating the Use of LLMs for Documentation to Code Traceability https://arxiv.org/html/2506.16440v1
48. LLM-Based Multi-Artifact Consistency Verification for Programming ... https://dl.acm.org/doi/10.1145/3769994.3770042
49. Structured Methods for LLM Consistency Checks - LinkedIn https://www.linkedin.com/top-content/innovation/process-improvement-methods/structured-methods-for-llm-consistency-checks/
50. [PDF] SCAR: Data Selection via Style Consistency-Aware Response ... https://aclanthology.org/2025.acl-long.625.pdf
51. KoLA - OpenDataLab https://opendatalab.com/OpenDataLab/KoLA/download
52. A High-Performance and Low-Energy Approach to PDF Parsing and ... https://arxiv.org/html/2602.00010v1
53. HTML Preprocessing for LLMs - DEV Community https://dev.to/rosgluk/html-preprocessing-for-llms-3mk8

54. Fine-grained Multilingual Benchmark Dataset for LLM Hallucinations https://arxiv.org/html/2503.07833v1
55. (PDF) HalluVerse25: Fine-grained Multilingual Benchmark Dataset ... https://www.researchgate.net/publication/389749153_HalluVerse25_Fine-grained_Multilingual_Benchmark_Dataset_for_LLM_Hallucinations
56. [PDF] Developing a Reliable, Fast, General-Purpose Hallucination ... https://aclanthology.org/2025.naacl-industry.72.pdf
57. [PDF] CHAIN-OF-VERIFICATION REDUCES HALLUCINATION https://openreview.net/pdf?id=ek3GqAs2uO
58. FED: Fast and Efficient Dataset Deduplication Framework with GPU ... https://arxiv.org/html/2501.01046v1
59. Data Preparation for Large Language Models - SciOpen https://www.sciopen.com/article/10.1007/s11390-026-5948-8
60. Artificial Intelligence - arXiv https://arxiv.org/list/cs.AI/new
61. Estimating Black-Box LLM Parameter Counts via Factual Capacity https://arxiv.org/html/2604.24827v1
62. How well LLM-based test generation techniques perform with newer ... https://arxiv.org/html/2601.09695v1
63. Step 3.5 Flash: Open Frontier-Level Intelligence with 11B Active ... https://arxiv.org/html/2602.10604v2
64. Large Language Models Hallucination: A Comprehensive Survey https://arxiv.org/html/2510.06265v2
65. The ARC of Progress towards AGI: A Living Survey of Abstraction ... https://arxiv.org/html/2603.13372v1
66. TokMem: One-Token Procedural Memory for Large Language Models https://arxiv.org/html/2510.00444v2
67. How to Interpret Agent Behavior - arXiv https://arxiv.org/html/2605.13625v1
68. Tongyi DeepResearch Technical Report - arXiv https://arxiv.org/html/2510.24701v1
69. Adversarial Self-Play with Evolving Distractions for LLM Reasoning https://arxiv.org/html/2605.11636v1
70. A Survey of Data Provenance, Transparency and Traceability in LLMs https://www.researchgate.net/publication/399965419_Tracing_the_Data_Trail_A_Survey_of_Data_Provenance_Transparency_and_Traceability_in_LLMs
71. Data × LLM: From Principles to Practices - arXiv https://arxiv.org/html/2505.18458v2

72. [PDF] AI Inferencing on Intel CPU-Powered Lenovo Servers https://lenovopress.lenovo.com/lp2148.pdf
73. [PDF] Compute Efficient Low-Memory LLM Training with Structured Sparse ... https://aclanthology.org/2024.emnlp-main.835.pdf
74. Full-Chain LLM Training | EvalScope https://evalscope.readthedocs.io/en/v0.7.1/best_practice/llm_full_stack.html
75. I Built an Open-Source Pipeline to Convert Documents into LLM ... https://dev.to/yevh/i-built-an-open-source-pipeline-to-convert-documents-into-llm-training-data-37pb
76. The Ultimate Guide to Fine-Tuning LLMs from Basics to Breakthroughs https://arxiv.org/html/2408.13296v1
77. Fine-Tuning Large Language Models to Improve Accuracy and ... https://dl.acm.org/doi/10.1145/3695993
78. LLM-Assisted Code Cleaning For Training Accurate Code Generators https://openreview.net/forum?id=maRYffiUpI
79. Data × LLM: From Principles to Practices - arXiv https://arxiv.org/html/2505.18458v3
80. Journal of Computer Research and Development 2026 Vol. 63 No. 6 https://crad.ict.ac.cn/en/article/current
81. LLM Quantization & Compression 2026 — PatSnap Eureka https://www.patsnap.com/resources/blog/rd-blog/llm-quantization-compression-2026-patsnap-eureka/
82. Vol. 40 No. 36: AAAI-26 Technical Tracks 36 https://ojs.aaai.org/index.php/AAAI/issue/view/718
83. GuidaPA: Privacy-Preserving Chatbot for Public Administration via ... https://arxiv.org/html/2606.01386v1
84. CivicShield: A Cross-Domain Defense-in-Depth Framework ... - arXiv https://arxiv.org/html/2603.29062v1
85. Umwelt Engineering: Designing the Cognitive Worlds of Linguistic ... https://arxiv.org/html/2603.27626v1
86. [PDF] Towards Excellent Comprehension of Public Policy for Large ... - arXiv https://arxiv.org/pdf/2604.12995
87. [PDF] Legal Alignment for Safe and Ethical AI - arXiv https://arxiv.org/pdf/2601.04175
88. [PDF] Interactive Evaluation Requires a Design Science - arXiv https://arxiv.org/pdf/2605.17829

89. [PDF] FinKario: Event-Enhanced Automated Construction of Financial ... https://arxiv.org/pdf/2508.00961
90. [PDF] From AI-Generated Content to Agentic Action: Security and Safety ... https://arxiv.org/pdf/2605.16471
91. [PDF] Hypothetical Terms Dataset for Benchmarking Hallucination ... https://aclanthology.org/2024.eacl-srw.9.pdf
92. Enhancing Large Language Model Reliability - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC11677479/
93. AI Inferencing on Intel CPU-Powered Lenovo Servers: Accelerating ... https://lenovopress.lenovo.com/lp2167-ai-inferencing-on-intel-cpu-powered-servers-accelerating-llm-performance
94. [PDF] Efficient LLM Inference on CPUs - Semantic Scholar https://www.semanticscholar.org/paper/Efficient-LLM-Inference-on-CPUs-Shen-Chang/c3460cb82948541e84635f3220fc60abd33afe8d
95. Unifying Datasets for Diverse, Effective Fine-tuning of LLM Agents https://arxiv.org/html/2510.24702v2
96. (PDF) Bridging the Data Provenance Gap Across Text, Speech and ... https://www.researchgate.net/publication/387382057_Bridging_the_Data_Provenance_Gap_Across_Text_Speech_and_Video
97. (PDF) The Common Pile v0.1: An 8TB Dataset of Public Domain and ... https://www.researchgate.net/publication/392466046_The_Common_Pile_v01_An_8TB_Dataset_of_Public_Domain_and_Openly_Licensed_Text
98. Large Language Model Prompt Datasets: An In-depth Analysis and ... https://arxiv.org/html/2510.09316v1
99. 1 Introduction - arXiv https://arxiv.org/html/2503.02784v2
100. ShortCoder: Knowledge-Augmented Syntax Optimization for Token ... https://arxiv.org/html/2601.09703v1
101. Running LLM DeepSeek on Raspberry Pi! I never ... - LinkedIn https://www.linkedin.com/posts/tanzirmehedi_running-llm-deepseek-on-raspberry-pi-i-activity-7290297801563947008-qvLa
102. [PDF] StarCoder2 and The Stack v2: The Next Generation - arXiv https://arxiv.org/pdf/2402.19173
103. StarCoder Training Dataset - 数据集详情页 https://modelscope.cn/datasets/AI-ModelScope/starcoderdata
104. Seed-Coder Open-Sourced: LLM-Based Code Data Building ... https://seed.bytedance.com/en/blog/seed-coder-open-sourced-llm-based-code-data-building-method-validated

105. LLM-Based Code Documentation Generation and Multi-Judge ... https://arxiv.org/html/2606.09852v1
106. Optimizing technical documentations for LLMs - DEV Community https://dev.to/joshtom/optimizing-technical-documentations-for-llms-4bcd
107. Verification Limits Code LLM Training | OpenReview https://openreview.net/forum?id=rkIwvPxArN
108. [PDF] Reliable Inline Code Documentation with LLMs: Fine-Grained ... https://aclanthology.org/2025.eval4nlp-1.4.pdf
109. [PDF] Hardware-Aware Parallel Prompt Decoding for Memory-Efficient ... https://aclanthology.org/2025.findings-emnlp.120.pdf
110. ONTO: A Token-Efficient Columnar Notation for LLM Input Optimization https://www.researchgate.net/publication/404021762_ONTO_A_Token-Efficient_Columnar_Notation_for_LLM_Input_Optimization
111. Bridging data and discovery: a survey on knowledge graphs in AI for ... https://academic.oup.com/nsr/article/13/8/nwag140/8507209
112. Datagraphy: toward a systematic approach to dataset discovery - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12644982/
113. [PDF] Automatic Identification of Research Articles related to Data ... https://aclanthology.org/2025.wasp-main.5.pdf
114. Reuse, Don't Retrain: A Recipe for Continued Pretraining of ... - arXiv https://arxiv.org/html/2407.07263v1
115. Continued Pretraining: A Practical Playbook for Language-Specific ... https://rocm.blogs.amd.com/artificial-intelligence/multilingual-continued-pretraining/README.html
116. Effectively doing Continued-Pretraining via Fine-tuning API - ChatGPT https://community.openai.com/t/effectively-doing-continued-pretraining-via-fine-tuning-api/918991
117. [PDF] Unleashing the Potential of Continuous Pretraining for Clinical LLMs. https://aclanthology.org/2024.findings-emnlp.618.pdf
118. PyAO: PyTorch-Based Memory-Efficient LLM Training on Ethernet ... https://www.mdpi.com/1424-8220/26/7/2269
119. Pie: Pooling CPU Memory for LLM Inference - arXiv https://arxiv.org/html/2411.09317v1
120. StarCoder 2 and The Stack v2: The Next Generation - 知乎专栏 https://zhuanlan.zhihu.com/p/32742509856
121. bigcode 发布StarCoder2 代码模型（3B、15B - 火山引擎开发者社区 https://developer.volcengine.com/articles/7390577392571383847

122. StarCoder 2 and The Stack v2: The Next Generation - 智源社区论文 https://hub.baai.ac.cn/paper/73309d08-78b6-4acd-8fe8-a718d0bd3717
123. starcoder2 - BigCode - Kaggle https://www.kaggle.com/models/bigcode-project/starcoder2/discussion/487804
124. [PDF] Accelerating On-Device LLM Inference via Activation-guided FFN ... https://openaccess.thecvf.com/content/CVPR2026W/MAI/papers/Moon_Accelerating_On-Device_LLM_Inference_via_Activation-guided_FFN_Distillation_on_Raspberry_CVPRW_2026_paper.pdf
125. A Standards-Focused Review of LLM-Based Assurance Techniques https://arxiv.org/html/2505.13766v2
126. LLM Testing Guide: 5 Strategies for 99% Accuracy - Testriq https://www.testriq.com/blog/post/llm-testing-guide-5-strategies-for-99-accuracy
127. Building a code review tool: The LLM patterns that actually work https://www.gresearch.com/news/building-a-code-review-tool-the-llm-patterns-that-actually-work/
128. Simplifying the Training of an LLM Model Using 1024 GPUs - MDPI https://www.mdpi.com/2076-3417/15/15/8265
129. [PDF] SciHallu: A Multi-Granularity Hallucination Detection Dataset for ... https://aclanthology.org/2025.ijcnlp-long.70.pdf
130. [PDF] provenance-driven data management for traceable scientific workflows https://hal.science/hal-03752208v1/file/mitchell-et-al-2022-fair-data-pipeline-provenance-driven-data-management-for-traceable-scientific-workflows.pdf
131. A Data Provenance Framework for Generative AI Datasets https://www.researchgate.net/publication/396942160_Compliance_Rating_Scheme_A_Data_Provenance_Framework_for_Generative_AI_Datasets
132. A study on formalizing the knowledge of data curation activities ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11045097/
133. Model-Agnostic Code Simplification for Pre-trained Large Language ... https://arxiv.org/abs/2405.11196
134. SwallowCode - 数据集详情页 https://www.modelscope.cn/datasets/AI-ModelScope/swallow-code
135. \emojidizzyStarCoder 2 and The Stack v2: The Next Generation - arXiv https://arxiv.org/html/2402.19173v1
136. the-stack-dedup - OpenDataLab https://opendatalab.com/OpenDataLab/the-stack-dedup
137. The Stack: 3 TB of permissively licensed source code - OpenReview https://openreview.net/forum?id=pxpbTdUEpD

138. (PDF) The Stack: 3 TB of permissively licensed source code https://www.researchgate.net/publication/365821542_The_Stack_3_TB_of_permissively_licensed_source_code
139. Enhance RAG performance through intelligent chunking strategies https://developer.ibm.com/articles/awb-enhancing-rag-performance-chunking-strategies/
140. [PDF] Structured Packing in LLM Training Improves Long Context Utilization https://ojs.aaai.org/index.php/AAAI/article/view/34706/36861
141. Compiled AI: Deterministic Code Generation for LLM-Based ... - arXiv https://arxiv.org/html/2604.05150v1
142. LLM generated Code Stylometry for Authorship Attribution https://dl.acm.org/doi/10.1145/3733799.3762964
143. Few-shot training LLMs for project-specific code-summarization https://www.researchgate.net/publication/366918308_Few-shot_training_LLMs_for_project-specific_code-summarization
144. [PDF] Generative AI for Technical Writing: Comparing Human and LLM ... https://aclanthology.org/2025.nodalida-1.67.pdf
145. Warning: AI Jargon is a Sign of Limited Understanding - LinkedIn https://www.linkedin.com/posts/glenmccracken_ive-been-having-a-giggle-lately-about-some-activity-7454424318509318144-xXf_
146. A Six-Layer Durable Write Surface for LLM Coding Agents - arXiv https://arxiv.org/html/2604.10842v3
147. Model-Agnostic Code Simplification for Pre-trained Large Language ... https://www.semanticscholar.org/paper/Natural-Is-the-Best%3A-Model-Agnostic-Code-for-Large-Wang-Li/fa3a28f092e2b49839d29b41b88e920bd06076d5
148. Model-Agnostic Code Simplification for Pre-trained Large Language ... https://www.researchgate.net/publication/380731035_Natural_Is_The_Best_Model-Agnostic_Code_Simplification_for_Pre-trained_Large_Language_Models
149. LEANCODE: Understanding Models Better for Code Simplification of ... https://aclanthology.org/2025.acl-long.78/
150. The Largest Collection of Ethical Data for LLM Pre-Training - arXiv https://arxiv.org/html/2506.01732v2
151. COIG - OpenDataLab https://opendatalab.com/OpenDataLab/COIG
152. Improving LLM Code Generation via Requirement-Aware ... - arXiv https://arxiv.org/html/2605.00433v1
153. Formal requirements engineering and large language models: A two ... https://www.sciencedirect.com/science/article/pii/S0950584925000369

154. (PDF) Verification Limits Code LLM Training - ResearchGate https://www.researchgate.net/publication/395848437_Verification_Limits_Code_LLM_Training
155. AI performance research papers - Red Hat https://www.redhat.com/en/artificial-intelligence/research
156. Pre-training of language models is possible even with HTML data! https://ai-scholar.tech/en/articles/natural-language-processing/HTLM
157. Boost LLM Accuracy with Clean Web Content | Divakar R posted on ... https://www.linkedin.com/posts/divakar01_ai-llm-rag-activity-7432502151253331968-UB4Y
158. [PDF] LEANCODE: Understanding Models Better for Code Simplification of ... https://aclanthology.org/2025.acl-long.78.pdf